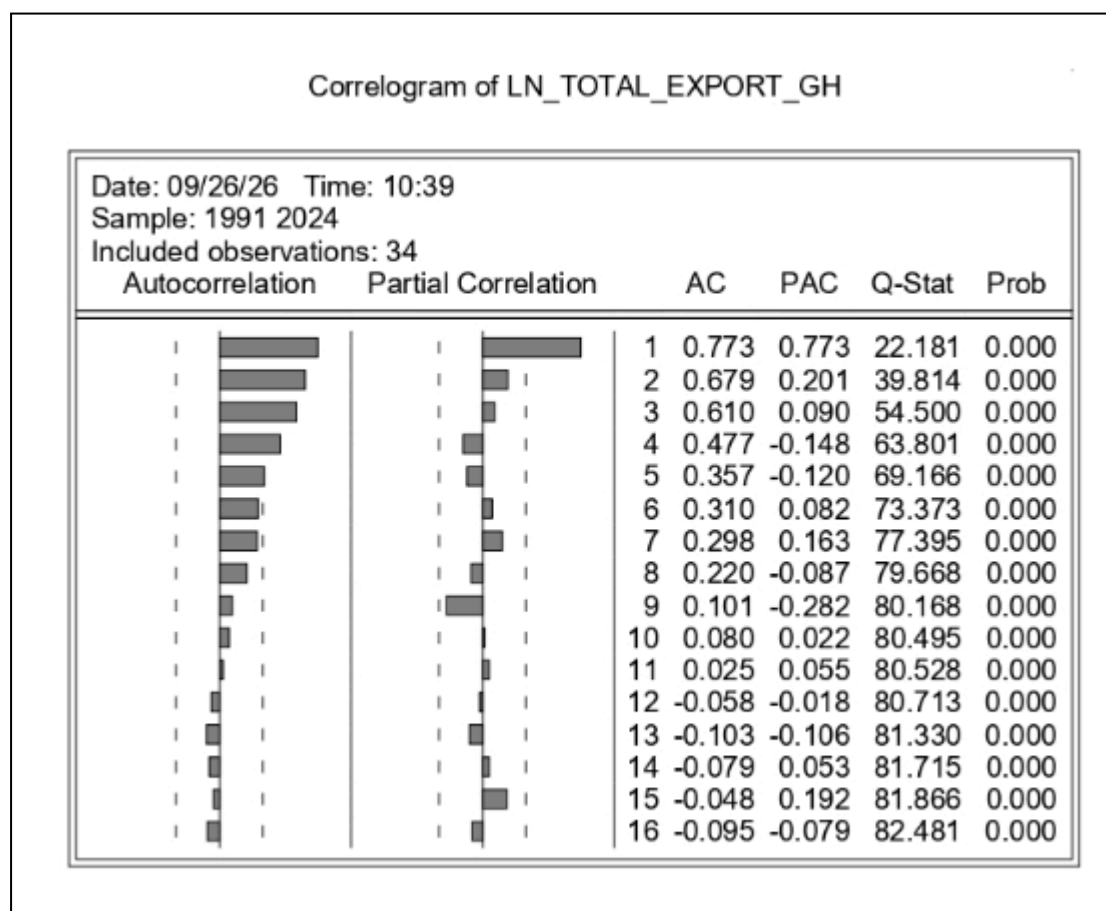
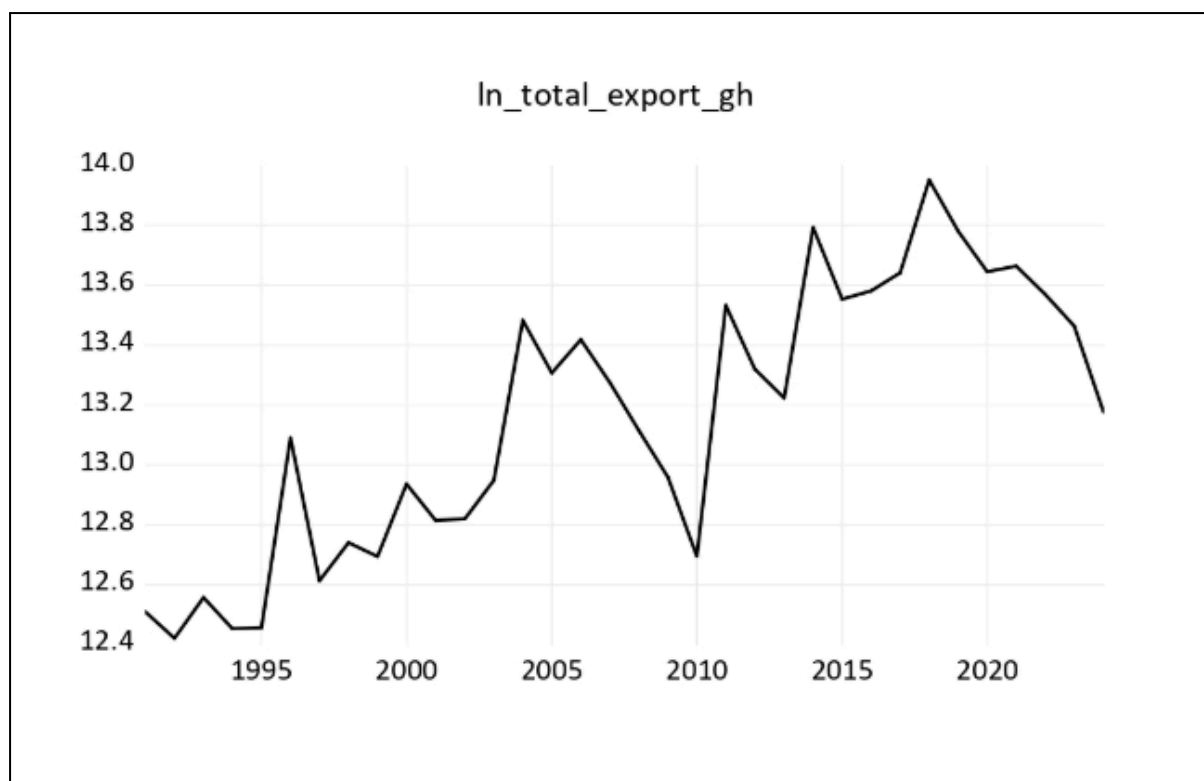
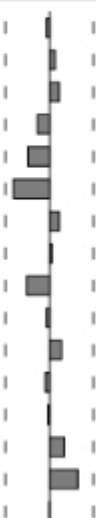
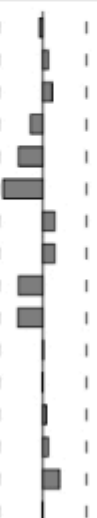




Debido a que la Función de Autocorrelaciones tarda varios desfases en anularse, y la Función de Autocorrelaciones Parciales se anula abruptamente en el segundo desfase, planteo inicialmente un modelo AR(1)

Dependent Variable: LN_TOTAL_EXPORT_GH				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 09/26/26 Time: 10:40				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Convergence achieved after 2 iterations				
Coefficient covariance computed using outer product of gradients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	12.55200	0.135572	92.58538	0.0000
@TREND	0.036049	0.006906	5.220176	0.0000
AR(1)	0.326757	0.195394	1.672295	0.1049
R-squared	0.706174	Mean dependent var		13.17251
Adjusted R-squared	0.686586	S.D. dependent var		0.443150
S.E. of regression	0.248091	Akaike info criterion		0.136462
Sum squared resid	1.846468	Schwarz criterion		0.272508
Log likelihood	0.748375	Hannan-Quinn criter.		0.182238
F-statistic	36.05065	Durbin-Watson stat		1.901546
Prob(F-statistic)	0.000000			
Inverted AR Roots	.33			

Correlogram of Residuals

Date: 09/26/26 Time: 10:58					
Sample (adjusted): 1992 2024					
Q-statistic probabilities adjusted for 1 ARMA term					
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 -0.021	-0.021	0.0160	
		2 0.045	0.045	0.0928	0.761
		3 0.078	0.080	0.3263	0.849
		4 -0.096	-0.096	0.6951	0.874
		5 -0.169	-0.183	1.8667	0.760
		6 -0.285	-0.306	5.3480	0.375
		7 0.079	0.091	5.6240	0.467
		8 0.019	0.096	5.6410	0.582
		9 -0.175	-0.181	7.1192	0.524
		10 -0.023	-0.190	7.1461	0.622
		11 0.097	0.012	7.6377	0.664
		12 -0.028	-0.006	7.6816	0.742
		13 -0.015	0.027	7.6940	0.809
		14 0.119	0.046	8.5543	0.806
		15 0.227	0.134	11.854	0.618
		16 -0.007	-0.000	11.858	0.690
*Probabilities may not be valid for this equation specification.					

Se ve como la tendencia con coeficiente positivo, y el coeficiente (ϕ_o) dieron significativos, sin embargo el coeficiente del AR(1) (ϕ_1) no dio significativo al 10%.

Dependent Variable: LN TOTAL EXPORT GH				
Method: Least Squares				
Date: 09/26/26 Time: 19:49				
Sample: 1991 2024				
Included observations: 34				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	12.52824	0.084289	148.6342	0.0000
@TREND	0.037865	0.004391	8.623567	0.0000
R-squared	0.699152	Mean dependent var	13.15301	
Adjusted R-squared	0.689750	S.D. dependent var	0.450959	
S.E. of regression	0.251184	Akaike info criterion	0.131762	
Sum squared resid	2.018993	Schwarz criterion	0.221548	
Log likelihood	-0.239962	Hannan-Quinn criter.	0.162382	
F-statistic	74.36591	Durbin-Watson stat	1.299920	
Prob(F-statistic)	0.000000			

Date: 09/26/26 Time: 19:50						
Sample: 1991 2024						
Included observations: 34						
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1 0.261	0.261	2.5177	0.113	
		2 0.097	0.032	2.8810	0.237	
		3 0.029	-0.004	2.9145	0.405	
		4 -0.163	-0.185	4.0012	0.406	
		5 -0.299	-0.239	7.7830	0.169	
		6 -0.325	-0.215	12.398	0.054	
		7 -0.008	0.171	12.401	0.088	
		8 -0.023	-0.018	12.425	0.133	
		9 -0.154	-0.254	13.589	0.138	
		10 -0.027	-0.136	13.626	0.191	
		11 0.125	0.101	14.452	0.209	
		12 0.021	-0.005	14.478	0.271	
		13 0.048	0.035	14.612	0.332	
		14 0.214	0.089	17.423	0.234	
		15 0.199	0.041	19.976	0.173	
		16 -0.026	-0.090	20.021	0.219	

Eliminando el componente AR(1) tanto la constante como la tendencia siguen siendo significativos.

Y al analizar los residuos vemos que se comportan como ruido blanco, lo que podría indicar que tiene tendencia determinística.

Sin embargo hay que realizar el test de Dickey-Fuller Ampliado para descartar la posibilidad de que tenga raíz unitaria.

Null Hypothesis: LN TOTAL EXPORT GH has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-3.445559	0.0625
Test critical values:			1% level	-4.262735
			5% level	-3.552973
			10% level	-3.209642
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN TOTAL EXPORT GH)				
Method: Least Squares				
Date: 09/26/26 Time: 19:47				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN TOTAL EXPORT GH(-1)	-0.673243	0.195394	-3.445559	0.0017
C	8.462324	2.434370	3.476186	0.0016
@TREND("1991")	0.024270	0.009253	2.622779	0.0136
R-squared	0.293695	Mean dependent var		0.020227
Adjusted R-squared	0.246608	S.D. dependent var		0.285825
S.E. of regression	0.248091	Akaike info criterion		0.136462
Sum squared resid	1.846468	Schwarz criterion		0.272508
Log likelihood	0.748375	Hannan-Quinn criter.		0.182238
F-statistic	6.237285	Durbin-Watson stat		1.901546
Prob(F-statistic)	0.005431			

El primer ADF test analiza un modelo con tendencia e intercepto, y tiene un p-valor de 0,0625 por lo que rechazo la hipótesis nula al 10% de significatividad pero no al 5%.

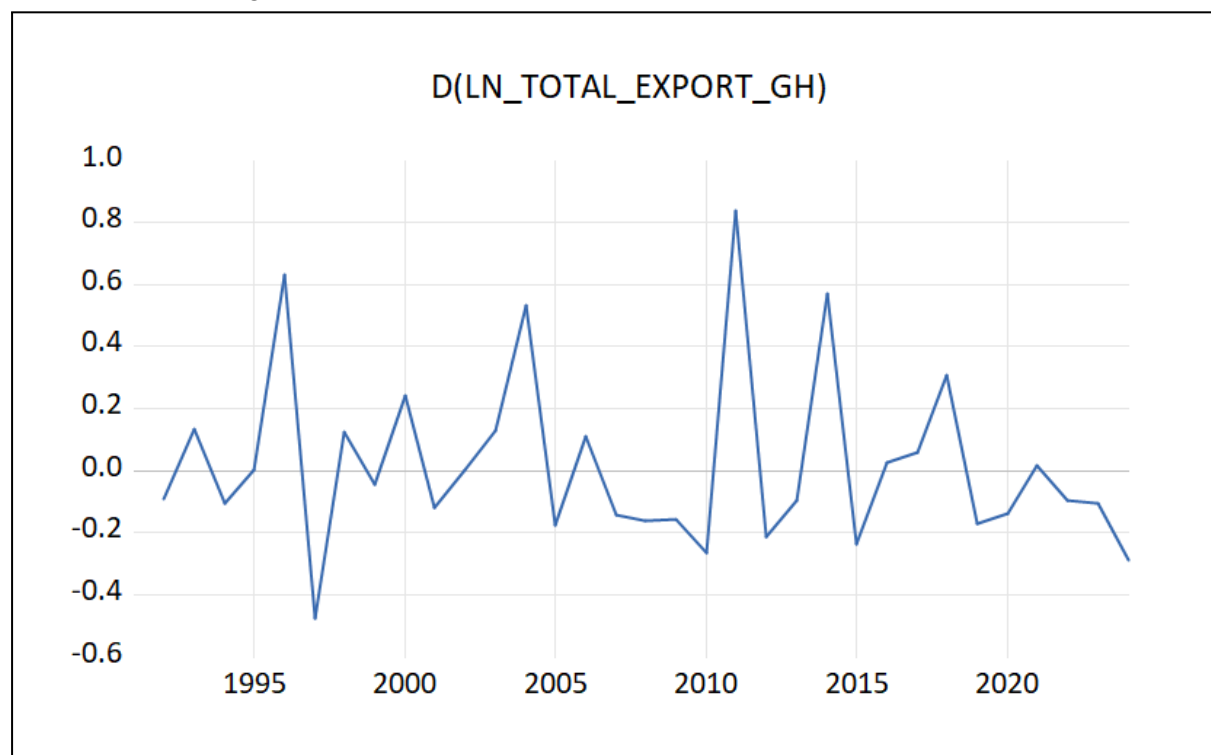
Null Hypothesis: LN_TOTAL_EXPORT_GH is stationary				
Exogenous: Constant, Linear Trend				
Bandwidth: 2 (Newey-West automatic) using Bartlett kernel				
				LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic				0.093456
Asymptotic critical values*:				1% level 0.216000
				5% level 0.146000
				10% level 0.119000
*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)				
Residual variance (no correction)				0.059382
HAC corrected variance (Bartlett kernel)				0.083869
KPSS Test Equation				
Dependent Variable: LN_TOTAL_EXPORT_GH				
Method: Least Squares				
Date: 09/27/26 Time: 14:59				
Sample: 1991 2024				
Included observations: 34				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	12.52824	0.084289	148.6342	0.0000
@TREND("1991")	0.037865	0.004391	8.623567	0.0000
R-squared	0.699152	Mean dependent var		13.15301
Adjusted R-squared	0.689750	S.D. dependent var		0.450959
S.E. of regression	0.251184	Akaike info criterion		0.131762
Sum squared resid	2.018993	Schwarz criterion		0.221548
Log likelihood	-0.239962	Hannan-Quinn criter.		0.162382
F-statistic	74.36591	Durbin-Watson stat		1.299920
Prob(F-statistic)	0.000000			

Aquí está el KPSS test que tiene como hipótesis nula que la serie es estacionaria. El resultado arroja que no se puede rechazar la H0 con un nivel de significancia del 10%. Por lo que teniendo en cuenta este test tengo dos resultados contradictorios entre sí, por un lado no rechazo que el modelo tenga raíz unitaria al 5%, pero si al 10%, por lo que habría que diferenciar la serie, y por otro no puedo rechazar que el modelo sea estacionario, por lo que al diferenciar el modelo podría estar sobrediferenciándolo.

Si considero que el ADF lo rechazo al 10% de significancia, y que además el KPSS me dio que es estacionario entonces el modelo quedó con tendencia e intercepto, y sin ningún componente AR/MA.

Si sigo con el nivel de significancia del 5% y que el modelo tiene raíz unitaria entonces debo diferenciar la serie una vez y obtengo:

$\Delta(\ln_total_export_gh)$



Date: 09/26/26 Time: 19:50

Sample (adjusted): 1992 2024

Included observations: 33 after adjustments

Included observations: 60 after adjustments						
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1	-0.372	-0.372	5.0054	0.025
		2	-0.040	-0.208	5.0660	0.079
		3	0.126	0.037	5.6782	0.128
		4	-0.038	0.029	5.7361	0.220
		5	-0.043	-0.029	5.8123	0.325
		6	-0.217	-0.315	7.8341	0.251
		7	0.195	-0.037	9.5199	0.217
		8	0.085	0.177	9.8539	0.275
		9	-0.197	-0.024	11.717	0.230
		10	0.046	-0.116	11.822	0.297
		11	0.073	-0.059	12.103	0.356
		12	-0.070	-0.058	12.372	0.416
		13	-0.050	-0.002	12.516	0.486
		14	-0.052	-0.111	12.681	0.552
		15	0.226	0.092	15.963	0.385
		16	-0.012	0.182	15.973	0.455

Se ve como tanto en la FAC como FACP solo el primer rezago es significativo por lo que podría ser un ARIMA(1,1,1).

Dependent Variable: D(LN TOTAL EXPORT GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 09/26/26 Time: 19:54				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Failure to improve likelihood (non-zero gradients) after 16 iterations				
Coefficient covariance computed using outer product of gradients				
MA Backcast: 1992				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.034451	0.011084	3.108003	0.0042
AR(1)	0.322743	0.209284	1.542130	0.1339
MA(1)	-0.928009	0.079549	-11.66582	0.0000
R-squared	0.248649	Mean dependent var		0.023628
Adjusted R-squared	0.196831	S.D. dependent var		0.289719
S.E. of regression	0.259645	Akaike info criterion		0.230060
Sum squared resid	1.955056	Schwarz criterion		0.367473
Log likelihood	-0.680958	Hannan-Quinn criter.		0.275608
F-statistic	4.798563	Durbin-Watson stat		1.901522
Prob(F-statistic)	0.015839			
Inverted AR Roots	.32			
Inverted MA Roots	.93			

Ahora la tendencia determinística deja de ser significativa por eso no está incluida en la estimación.

El coeficiente ϕ_1 del componente AR no es significativo por lo que lo retiro de la estimación.

Dependent Variable: D(LN TOTAL EXPORT GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 09/26/26 Time: 19:55				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Failure to improve likelihood (non-zero gradients) after 12 iterations				
Coefficient covariance computed using outer product of gradients				
MA Backcast: 1991				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.028921	0.022108	1.308168	0.2004
MA(1)	-0.540173	0.170880	-3.161116	0.0035
R-squared	0.188953	Mean dependent var		0.020227
Adjusted R-squared	0.162790	S.D. dependent var		0.285825
S.E. of regression	0.261527	Akaike info criterion		0.214135
Sum squared resid	2.120292	Schwarz criterion		0.304833
Log likelihood	-1.533234	Hannan-Quinn criter.		0.244652
F-statistic	7.222182	Durbin-Watson stat		1.842635
Prob(F-statistic)	0.011475			
Inverted MA Roots	.54			

Al sacar el componente AR la constante dejó de ser significativa por lo que también la retiro.

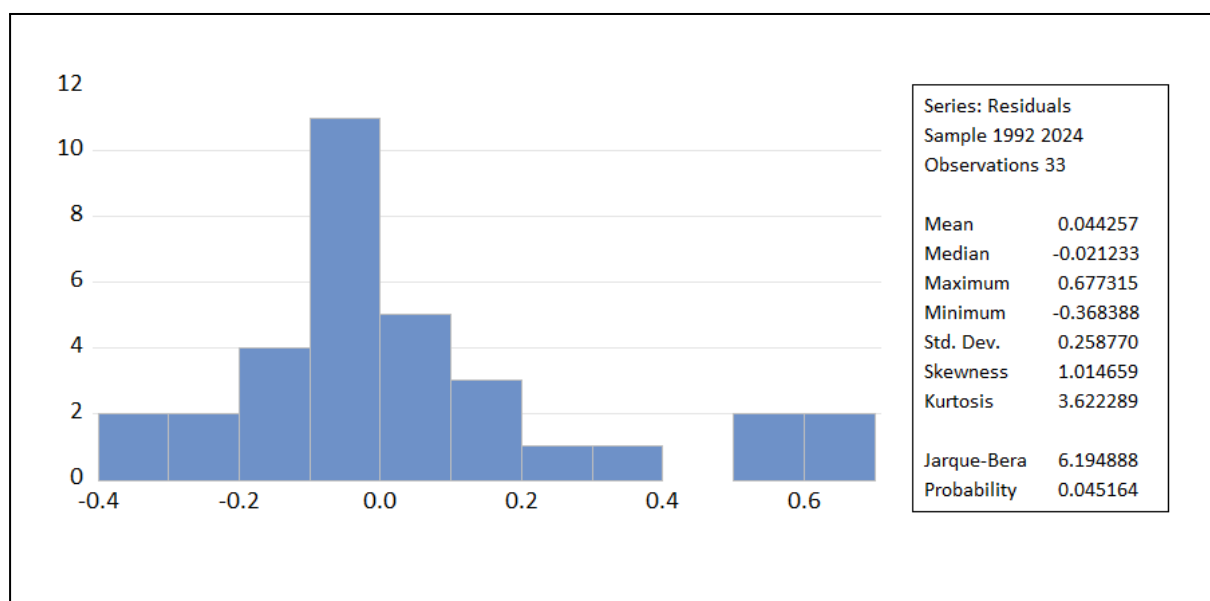
Dependent Variable: D(LN TOTAL EXPORT GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 09/26/26 Time: 19:56				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Failure to improve likelihood (non-zero gradients) after 7 iterations				
Coefficient covariance computed using outer product of gradients				
MA Backcast: 1991				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	-0.433615	0.167110	-2.594789	0.0142
R-squared	0.155625	Mean dependent var		0.020227
Adjusted R-squared	0.155625	S.D. dependent var		0.285825
S.E. of regression	0.262644	Akaike info criterion		0.193800
Sum squared resid	2.207420	Schwarz criterion		0.239149
Log likelihood	-2.197697	Hannan-Quinn criter.		0.209058
Durbin-Watson stat	1.969187			
Inverted MA Roots	.43			

El coeficiente restante MA(1) es significativo al 5% por lo que verifico los residuos.

Date: 09/26/26 Time: 19:56
Sample (adjusted): 1992 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.055 -0.055	0.1095		
		2 -0.042 -0.045	0.1757	0.675	
		3 0.077 0.072	0.4032	0.817	
		4 -0.077 -0.072	0.6415	0.887	
		5 -0.169 -0.173	1.8191	0.769	
		6 -0.241 -0.285	4.3104	0.506	
		7 0.161 0.130	5.4673	0.485	
		8 0.083 0.122	5.7892	0.565	
		9 -0.173 -0.160	7.2335	0.512	
		10 -0.006 -0.156	7.2352	0.613	
		11 0.080 -0.014	7.5719	0.671	
		12 -0.079 -0.036	7.9195	0.720	
		13 -0.071 0.008	8.2079	0.769	
		14 0.066 -0.000	8.4746	0.811	
		15 0.245 0.159	12.322	0.580	
		16 0.017 0.077	12.340	0.653	

Los residuos del modelo ARIMA(0,1,1) sin constante se comportan como ruido blanco.
Ahora verificamos con el test de normalidad.



Con un nivel de significancia del 5% el test de Jarque-Bera rechaza la hipótesis nula de que los residuos se comportan de forma Normal, debido a la presencia de una asimetría positiva.

Null Hypothesis: D(LN_TOTAL_EXPORT_GH) has a unit root				
Exogenous: None				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-8.112022	0.0000
Test critical values:	1% level		-2.639210	
	5% level		-1.951687	
	10% level		-1.610579	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN_TOTAL_EXPORT_GH,2)				
Method: Least Squares				
Date: 09/27/26 Time: 15:16				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_TOTAL_EXPORT_GH(-1))	-1.374019	0.169381	-8.112022	0.0000
R-squared	0.679713	Mean dependent var		-0.006200
Adjusted R-squared	0.679713	S.D. dependent var		0.477499
S.E. of regression	0.270235	Akaike info criterion		0.251704
Sum squared resid	2.263841	Schwarz criterion		0.297508
Log likelihood	-3.027263	Hannan-Quinn criter.		0.266887
Durbin-Watson stat	2.089343			

Con el test de Dickey-Fuller sin intercepto ni tendencia se confirma con holgura (0.01%) que el modelo en diferencias no posee ninguna raíz unitaria.

Dependent Variable: D(LN_TOTAL_EXPORT_GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 09/27/26 Time: 15:10				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Convergence achieved after 3 iterations				
Coefficient covariance computed using outer product of gradients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	-0.374019	0.169381	-2.208156	0.0348
R-squared	0.129979	Mean dependent var		0.023628
Adjusted R-squared	0.129979	S.D. dependent var		0.289719
S.E. of regression	0.270235	Akaike info criterion		0.251704
Sum squared resid	2.263841	Schwarz criterion		0.297508
Log likelihood	-3.027263	Hannan-Quinn criter.		0.266887
Durbin-Watson stat	2.089343			
Inverted AR Roots	-.37			

Date: 09/27/26 Time: 15:11					
Sample (adjusted): 1993 2024					
Q-statistic probabilities adjusted for 1 ARMA term					
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.093	-0.093	0.3050	
		2 -0.177	-0.188	1.4452	0.229
		3 0.122	0.089	2.0071	0.367
		4 0.011	-0.001	2.0116	0.570
		5 -0.198	-0.168	3.5845	0.465
		6 -0.236	-0.302	5.9151	0.315
		7 0.215	0.109	7.9192	0.244
		8 0.123	0.134	8.6057	0.282
		9 -0.216	-0.117	10.809	0.213
		10 -0.002	-0.113	10.809	0.289
		11 0.117	-0.048	11.524	0.318
		12 -0.067	-0.026	11.766	0.381
		13 -0.116	0.004	12.540	0.403
		14 0.048	-0.023	12.679	0.473
		15 0.251	0.145	16.726	0.271
		16 -0.020	0.076	16.754	0.334

Modelando un ARIMA(1,1,0) también hubiera llegado a ruido blanco, sin embargo utilizo el criterio de Schwarz para decidir entre modelos y como el modelo ARIMA(0,1,1) tiene un valor más pequeño es el modelo que utilizo ($0.23 < 0.29$).

De este modo queda modelada la variable $\ln_total_export_gh$, que representa la oferta de cacao de Ghana, formada por la exportación de cacao y de todos sus derivados debidamente ponderados:

- Como un ARIMA(0,1,1), si se sigue a rajatabla el ADF test con el nivel de significancia del 5%.
- Como un modelo sin componentes AR/MA con intercepto y tendencia determinística significativos, si se considera que el ADF test es insuficiente y se utiliza el KPSS test que indica que la serie es estacionaria en torno a la tendencia determinística.